CodeUI Documentation

CodeUI is a fluent, declarative C# framework for building user interfaces with Unity UI Toolkit (UIElements). It removes the boilerplate associated with UXML and USS by enabling full layout, styling, event binding, and state transitions directly in code.

Features

- **Fluent API**: Chain extension methods to style, layout, and position visual elements directly in C#.
- **Design Tokens**: Standardized constants for spacing (`Spacing`), font sizes (`FontSize`), and corner radii (`Radius`).
- **Unity Data Binding**: Declarative wrappers for binding properties to Unity Data Sources via `BindText` and `BindValue`.
- **Type-Safe Transitions**: Animate CSS properties cleanly using the `TransitionProperty` enum.
- **Zero External Dependencies**: Built natively on top of `UnityEngine.UIElements`.

Quick Example

Inherit from `BaseUI` to define your component hierarchy:

```
using CodeUI;
using CodeUI.Extensions;
using CodeUI.Tokens;
using UnityEngine;
using UnityEngine.UIElements;

public class MainMenuUI : BaseUI
{
    protected override void Generate(VisualElement root)
    {
        root.Column()
            .Center()
            .Padding(Spacing.LG)
            .WithChildren(
                new Label("Main Menu")
                    .FontSize(FontSize.XL)
                    .Margin(bottom: Spacing.MD),

                new Button(() => Debug.Log("Start Game"))
                    .SetText("Play")
                    .Padding(Spacing.SM, Spacing.MD)
                    .BorderRadius(Radius.MD)
            );
```

```
    }
}
```

Documentation

- Getting Started Guide
- API Reference

Getting Started

CodeUI provides a fluent C# interface for constructing Unity UI Toolkit components.

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1. Creating Components with BaseUI

Attach a `UIDocument` component to a GameObject and inherit from `BaseUI` to construct your interface inside `Generate`:

```
using CodeUI;
using CodeUI.Extensions;
using CodeUI.Tokens;
using UnityEngine;
using UnityEngine.UIElements;

public class SettingsScreen : BaseUI
{
    protected override void Generate(VisualElement root)
    {
        root.Column()
            .Padding(Spacing.MD)
            .WithChildren(
                new Label("Settings").FontSize(FontSize.LG),

                new Toggle("Mute Audio")
                    .OnValueChanged<Toggle, bool>(evt => AudioListener.pause
= evt.newValue),

                Spacer.Vertical(Spacing.MD),

                new Button(() => RebuildUI())
                    .SetText("Refresh")
            );
    }
}
```

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2. Layout & Positioning

CodeUI provides shorthand extension methods for standard Flexbox properties:

Method	Description
`Row()`	Sets flex direction to horizontal (`FlexDirection.Row`)
`Column()`	Sets flex direction to vertical (`FlexDirection.Column`)
`Center()`	Centers elements horizontally and vertically (`Align.Center`, `Justify.Center`)
`Absolute()`	Sets position mode to absolute (`Position.Absolute`)
`Relative()`	Sets position mode to relative (`Position.Relative`)

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3. Styling & Tokens

Standard design tokens ensure consistent spacing, typography, and layout boundaries across components:

- **Spacing**: `Spacing.XS` (4px), `Spacing.SM` (8px), `Spacing.MD` (16px), `Spacing.LG` (24px), `Spacing.XL` (32px), `Spacing.XXL` (48px)
- **FontSize**: `FontSize.XS` (12px), `FontSize.SM` (14px), `FontSize.MD` (16px), `FontSize.LG` (20px), `FontSize.XL` (28px), `FontSize.XXL` (40px)
- **Radius**: `Radius.None` (0px), `Radius.SM` (4px), `Radius.MD` (8px), `Radius.LG` (16px), `Radius.Full` (999px)

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4. Event Handling

Events can be attached directly to visual elements during instantiation:

```
button.OnClick(() => Debug.Log("Button clicked"))
      .OnHover(isHovering => button.ToggleClasses(isHovering, "hovered"));
```

---

5. Animations & Transitions

Smooth CSS transitions can be defined using the type-safe `TransitionProperty` enum:

```
card.Transition(TransitionProperty.scale, durationSeconds: 0.2f, EasingMode.EaseOutCubic)
    .OnHover(hovered => card.Scale(hovered ? 1.05f : 1.0f));
```

Namespace CodeUI

Classes

BaseUI

BaseUIEditor